S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Case Study Track

SENTINELRISK

Unified AI/NLP Financial Risk Engine & Real-Time Capital Analytics

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github.com/Aayushi002929/financial-risk-engine
financial-risk-engine-2.vercel.app

The Signal Gap: risk models lag the narrative



Information asymmetry

News wires, social feeds and SEC filings overwhelm manual analysis.



Lagging indicators

Models rely on backward-looking volatility and end-of-day closes.



Execution gap

Sentiment tools sit in silos, disconnected from portfolio rebalancing and banking books.

SentinelRisk turns unstructured financial narrative into machine-readable risk scores and immediate capital actions.

One signal, two decisions

01

Ingest

Bloomberg/Reuters wires, X/Twitter feeds, custom analyst inputs.

02

Interpret

Sentiment (-1.0 to +1.0), 6 event categories, severity (1 to 10), target ticker.

03

Execute

Module A

Rebalances an S&P 10-stock basket with exact 100.00% normalization.

Module B

Stress-tests a $140M wholesale credit book, auto-triggered at Impact >= 7.

System design: fast, deterministic, zero-dependency

INGESTION LAYER

Bloomberg, Reuters, WSJ, X/Twitter stream, custom workbench

PROCESSING CORE

Entity normalization, calibrated financial lexicon sentiment, multi-class classification, non-linear severity scoring

EXECUTION

Module A

Delta W_i = Sentiment * 3.5%

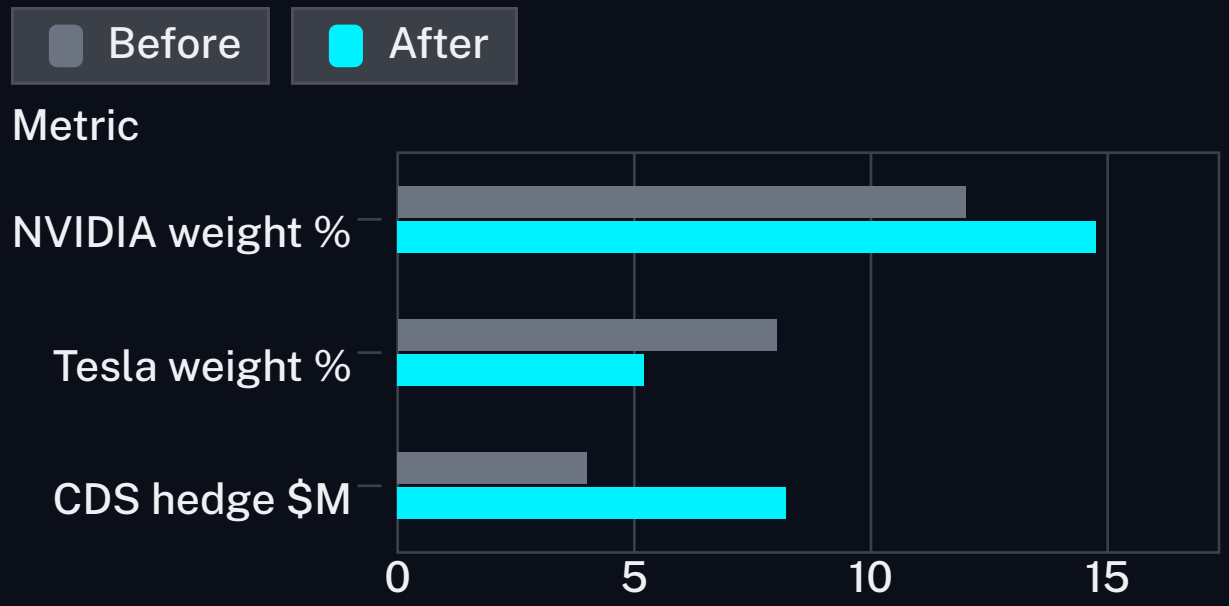
Strict sum(W) = 100.00%

Module B

Credit spread widening, duration sensitivity, non-linear CDS hedge appreciation

ES6+ and Python standard library · under 5ms latency · TailwindCSS · Chart.js · Vercel edge

Prototype readout: gains, pruning and a hedged drawdown



NVIDIA sentiment +0.79: 12.00% to 14.76% (+2.76%)
Tesla sentiment -0.80: 8.00% to 5.20% (-2.80%)

Basket exact 100.00% normalization maintained

GEOPOLITICAL SHOCK

Severity 10 · +180 bps spread

$130.82M

$140M book fell -6.56%
$9.18M loss

CDS hedge +105%

$4.0M to $8.2M, buffering loan losses

Vs traditional models: real-time under 5ms vs days · automated triggers vs quarterly review · non-linear vs static linear response

Institutional value for buy-side and sell-side

Asset management

Captures narrative momentum before earnings revisions; cuts exposure to legally impaired stocks.

Wholesale banking

Early-warning surveillance on syndicated loans ahead of rating downgrades; continuous Basel III capital testing.

For S&P Global & CRISIL

Augments CreditModel and Capital IQ credit surveillance; enables sentiment-adjusted smart-beta index products.



Roadmap: from prototype to enterprise scale

01

Today

Domain lexicon favors sub-5ms deterministic speed.

Modified duration uses linear approximations.

02

Next release

Hybrid LLM / FinBERT reasoning for multi-lingual filings and complex disclosures.

03

Enterprise scale

- Live Apache Kafka and SEC EDGAR ingestion
- Monte Carlo VaR across 10,000+ facilities
- S&P Capital IQ API integration

Thank you!

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